Exercises: Topological Basics and Continuity

Mathematical Foundations for Machine Learning

Difficulty: $\star$ Easy $\star\star$ Medium $\star\star\star$ Hard

1 Open and Closed Sets

Exercise 1.1 ($\star$). *Let $\Omega = (0,1) \subset \mathbb{R}$. Using the definition of an open set, prove that Ω is open. That is, show that for every $x \in (0,1)$ there exists $\varepsilon > 0$ such that $B(x,\varepsilon) \subset (0,1)$.*

Exercise 1.2 ($\star$). *Determine whether the following subsets of $\mathbb{R}^2$ (with the Euclidean norm) are open, closed, both, or neither. Justify each answer.*

(a) $A = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$

(b) $B = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$

(c) $C = \{(x,y) \in \mathbb{R}^2 : 0 < x^2 + y^2 \leq 1\}$

(d) $D = \mathbb{R}^2$

Exercise 1.3 ($\star\star$). *Prove that the set $\Omega = \{(x,y) \in \mathbb{R}^2 : x > 0 \text{ and } y > 0\}$ (the open first quadrant) is an open set using the definition with open balls.*

Exercise 1.4 ($\star\star$). *Let $\Omega = [0,1] \subset \mathbb{R}$. Using the sequential characterization of closed sets (every convergent sequence with elements in Ω has its limit in Ω), prove that Ω is closed.*

Exercise 1.5 ($\star\star\star$). *Prove that a finite intersection of open sets is open, and that an arbitrary union of open sets is open. Give an example showing that an infinite intersection of open sets need not be open.*

2 Closure, Interior, and Boundary

Exercise 2.1 ($\star$). *For the set $\Omega = (0,1] \subset \mathbb{R}$, determine:*

(a) The closure $\overline{\Omega}$

(b) The interior Ω^o

(c) The boundary $\partial\Omega$

Verify that $\partial\Omega = \overline{\Omega} \setminus \Omega^o$.

Exercise 2.2 ($\star$). *For $\Omega = \mathbb{Q} \cap [0,1]$ (the rationals in $[0,1]$), find $\overline{\Omega}$, Ω^o, and $\partial\Omega$.*

Exercise 2.3 ($\star\star$). *Let $\Omega = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1\} \cup \{(1,0)\}$. Find the closure, interior, and boundary of Ω.*

Exercise 2.4 ($\star\star$). *Prove that for any set $\Omega \subset \mathbb{R}^n$:*

(a) Ω^o *is open.*

(b) $\overline{\Omega}$ *is closed.*

(c) Ω *is open if and only if* $\Omega = \Omega^o$.

Exercise 2.5 ($\star\star\star$). *Let $A, B \subset \mathbb{R}^n$. Prove or give a counterexample:*

(a) $\overline{A \cup B} = \overline{A} \cup \overline{B}$

(b) $\overline{A \cap B} = \overline{A} \cap \overline{B}$

(c) $(A \cap B)^o = A^o \cap B^o$

(d) $(A \cup B)^o = A^o \cup B^o$

3 Boundedness, Compactness, and Bolzano–Weierstrass

Exercise 3.1 ($\star$). *Determine which of the following subsets of $\mathbb{R}^n$ are bounded. Justify your answers.*

(a) $A = \{x \in \mathbb{R}^2 : \|x\|_2 \leq 5\}$

(b) $B = \{x \in \mathbb{R}^2 : x_1^2 - x_2^2 < 1\}$

(c) $C = \{x \in \mathbb{R}^n : \|x\|_1 < 10\}$

Exercise 3.2 ($\star$). *Determine which of the following sets are compact in $\mathbb{R}^n$. Justify using Theorem 52 (compact $\Leftrightarrow$ closed and bounded in $\mathbb{R}^n$).*

(a) $[0,1]^n$

(b) $(0,1)^n$

(c) $\{x \in \mathbb{R}^n : \|x\|_2 = 1\}$ *(the unit sphere)*

(d) $\{1/n : n \in \mathbb{N}\}$

(e) $\{0\} \cup \{1/n : n \in \mathbb{N}\}$

Exercise 3.3 ($\star\star$). *Let $(x_k)_{k\in\mathbb{N}}$ be a sequence in $\mathbb{R}^2$ defined by $x_k = \left(\cos(k), \frac{(-1)^k}{k}\right)$. Show that this sequence is bounded and use the Bolzano–Weierstrass theorem to conclude that it has a convergent subsequence.*

Exercise 3.4 ($\star\star$). *Let $K \subset \mathbb{R}^n$ be compact and let (x_k) be a sequence in K such that every convergent subsequence converges to the same limit $x^* \in K$. Prove that the full sequence (x_k) converges to x^*.*

Exercise 3.5 ($\star\star\star$). *Let $K_1 \supset K_2 \supset K_3 \supset \cdots$ be a nested sequence of non-empty compact sets in $\mathbb{R}^n$. Prove that $\bigcap_{i=1}^{\infty} K_i \neq \emptyset$. Show by example that the conclusion can fail if the sets are only closed (but not bounded).*

4 Continuity

Exercise 4.1 ($\star$). *Using the ε-δ definition, prove that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 3x + 2$ is continuous.*

Exercise 4.2 ($\star$). *Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(x) = \|x\|_2$. Prove that f is continuous. Hint: use the reverse triangle inequality.*

Exercise 4.3 ($\star\star$). *Prove using the preimage characterization (Proposition 3) that the set $\{x \in \mathbb{R}^n : \|x\|_2 < 1\}$ is open and the set $\{x \in \mathbb{R}^n : \|x\|_2 \leq 1\}$ is closed.*

Exercise 4.4 ($\star\star$). *Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be continuous and let $U \subset \mathbb{R}^n$ be open. Is $f(U)$ necessarily open? Prove or give a counterexample.*

Exercise 4.5 ($\star\star\star$). *Let $f : K \to \mathbb{R}$ be continuous on a compact set $K \subset \mathbb{R}^n$. Prove that f is uniformly continuous on K, i.e., for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|f(x) - f(y)\| < \varepsilon$ for all $x, y \in K$ with $\|x - y\| < \delta$.*

5 Extreme Value Theorem and Distance to Sets

Exercise 5.1 ($\star$). *Let $f(x) = x^2 - 2x + 3$ on $K = [0, 3]$. Find the minimum and maximum of f on K and state which theorem guarantees their existence.*

Exercise 5.2 ($\star\star$). *Let $K = \{x \in \mathbb{R}^2 : \|x\|_2 \leq 1\}$ and $f(x) = x_1^2 + 2x_2^2 - x_1$. Find the minimum and maximum of f on K and verify that Corollary 7 applies.*

Exercise 5.3 ($\star\star$). *Let $K = \{x \in \mathbb{R}^2 : x_1 \geq 0,\ x_2 \geq 0,\ x_1 + x_2 \leq 1\}$ (a triangle). Show that K is compact, and for $p = (2, 2)$, find $d(p, K) = \inf_{u \in K} \|p - u\|_2$ and the point in K that attains it.*

Exercise 5.4 ($\star\star\star$). *Let $M \subset \mathbb{R}^n$ be a non-empty closed set and $x \in \mathbb{R}^n$. Prove that there exists $m^* \in M$ such that $\|x - m^*\| = d(x, M) := \inf_{u \in M} \|x - u\|$. Hint: reduce to optimization over a compact set as described in the notes.*

6 Path-Connectedness

Exercise 6.1 ($\star$). *Show that every convex set $C \subset \mathbb{R}^n$ is path-connected. Recall that C is convex if for all $x, y \in C$ and $t \in [0, 1]$, $(1 - t)x + ty \in C$.*

Exercise 6.2 ($\star\star$). *Show that the set $\Omega = \mathbb{R}^2 \setminus \{(0, 0)\}$ is path-connected.*

Exercise 6.3 ($\star\star$). *Prove that if $f : V_1 \to V_2$ is continuous and $M \subset V_1$ is path-connected, then $f(M)$ is path-connected.*

Exercise 6.4 ($\star\star\star$). *Consider the set of invertible $n \times n$ real matrices $\mathrm{GL}_n(\mathbb{R}) = \{A \in \mathbb{R}^{n \times n} : \det(A) \neq 0\}$.*

(a) Prove that $\mathrm{GL}_n(\mathbb{R})$ is an open set in $\mathbb{R}^{n \times n}$.

(b) Prove that $\mathrm{GL}_n(\mathbb{R})$ has exactly two path-connected components: $\{A : \det(A) > 0\}$ and $\{A : \det(A) < 0\}$.

7 Mixed Challenging Exercises

Exercise 7.1 ($\star\star\star$). *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous. Prove that for every $c \in \mathbb{R}$, the sublevel set $S_c = \{x \in \mathbb{R}^n : f(x) \leq c\}$ is closed. Explain why this is relevant in optimization (e.g., for defining feasible sets).*

Exercise 7.2 ($\star\star\star$). *Let $K \subset \mathbb{R}^n$ be compact and non-empty. Prove that K has a finite diameter, i.e., $\operatorname{diam}(K) = \sup_{x,y \in K} \|x - y\| < \infty$, and that the supremum is attained.*

Exercise 7.3 ($\star\star\star$). *Let (f_k) be a sequence of continuous functions $f_k : K \to \mathbb{R}$ on a compact set $K \subset \mathbb{R}^n$ that converges uniformly to $f : K \to \mathbb{R}$. Prove that f is continuous. Provide a counterexample showing that pointwise convergence is not sufficient.*

Exercise 7.4 ($\star\star\star$). *Let $A \in \mathbb{R}^{m \times n}$ and define $f : \mathbb{R}^n \to \mathbb{R}^m$ by $f(x) = Ax$. Prove that:*

(a) f is continuous.

(b) If $C \subset \mathbb{R}^n$ is compact, then $f(C)$ is compact.

(c) If $M \subset \mathbb{R}^n$ is path-connected, then $f(M)$ is path-connected.

(d) The null space $\ker(A) = \{x : Ax = 0\}$ is closed.

Exercise 7.5 ($\star\star\star$). ***(Application to ML)*** *Consider a neural network layer $f(x) = \sigma(Wx+b)$ where σ is a continuous activation function (e.g., ReLU, sigmoid), $W \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^m$.*

(a) Prove that f is continuous.

(b) If the input is constrained to a compact set K (e.g., normalized pixel values in $[0, 1]^n$), what can you conclude about the range $f(K)$?

(c) Explain why path-connectedness of the loss landscape over weight space matters for gradient-based optimization.

Exercises: Differentiation in Several Variables

Mathematical Foundations for Machine Learning

Partial Derivatives

Exercise 1.1 [Easy]
Compute all partial derivatives of $f(x, y) = x^3y + 2xy^2 - 5y$.

Exercise 1.2 [Easy]
Let $f(x, y, z) = e^{xz}\sin(y)$. Compute $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, and $\frac{\partial f}{\partial z}$.

Exercise 1.3 [Medium]
Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \dfrac{x^2y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

the derivative is undefined because if you take lim h-> 0 (h, h) and lim h->0 (h,-h) its different limits, so depends on from where you approach it is different limits.

Compute $\frac{\partial f}{\partial x}(0, 0)$ and $\frac{\partial f}{\partial y}(0, 0)$ using the limit definition.

Exercise 1.4 [Medium]
Let $f(x, y) = \ln(x^2 + y^2)$ for $(x, y) \neq (0, 0)$. Use the quotient rule to compute both partial derivatives and simplify.

Gradients and Jacobians

Exercise 2.1 [Easy]
Compute the gradient of $f(x, y, z) = x^2y + yz^3 - 3xz$.

Exercise 2.2 [Easy]
Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $f(x, y) = \begin{pmatrix} x^2 + y \\ xy \end{pmatrix}$. Compute the Jacobian $Df(x, y)$.

Exercise 2.3 [Medium]
Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $f(x, y, z) = \begin{pmatrix} e^{x+y} \\ \sin(xz) \end{pmatrix}$. Compute the Jacobian $Df(x, y, z)$ and verify that each row is the transpose of the gradient of the corresponding component.

Exercise 2.4 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(x) = \langle a, x\rangle + b$ where $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$. Show that $\nabla f(x) = a$ for all x. What is the Jacobian?

Total Differentiability

Exercise 3.1 [Easy]
Let $f(x, y) = 3x + 2y - 1$. Show that f is totally differentiable at every point and identify the matrix A from the definition of total differentiability.

To be totally differentiable we need all the partials to be defined + R(h) vanish before ||h|| -> else, it could happ en like the example where the derivative was not defined but the partials existed.

Exercise 3.2 [Medium]
Consider the function

$$f(x, y) = \begin{cases} \dfrac{xy}{\sqrt{x^2 + y^2}} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Show that both partial derivatives exist at the origin, but f is **not** totally differentiable at $(0, 0)$.

Exercise 3.3 [Hard]
Prove: if $f : \mathbb{R}^n \to \mathbb{R}^m$ is totally differentiable at x, then f is continuous at x.

Hint: Use the definition $f(x + h) = f(x) + Ah + R(h)$ and the fact that $\|R(h)\| \leq \|h\|$ for $\|h\|$ small enough.

Chain Rule and Product Rule

Exercise 4.1 [Easy]
Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ with $f(x, y) = (x + y,\, xy)$ and $g : \mathbb{R}^2 \to \mathbb{R}$ with $g(u, v) = u^2 + v$. Compute $D(g \circ f)(x, y)$ using the chain rule.

Exercise 4.2 [Easy]
Let $f(x, y) = x^2$ and $g(x, y) = e^y$. Use the product rule to compute $\nabla(f \cdot g)(x, y)$.

Exercise 4.3 [Medium]
Let $\gamma(t) = (\cos t,\, \sin t)$ be a curve in $\mathbb{R}^2$ and $f(x, y) = x^2 + y^2$. Using the chain rule, compute $\frac{d}{dt} f(\gamma(t))$. Interpret your result geometrically.

Exercise 4.4 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}^m$ and $g : \mathbb{R}^m \to \mathbb{R}^l$ be totally differentiable. Using the definition of total differentiability, prove the chain rule:

$$D(g \circ f)(x) = Dg(f(x))\, Df(x).$$

Hessian and Schwarz's Theorem

Exercise 5.1 [Easy]
Compute the Hessian matrix $H_f(x, y)$ of $f(x, y) = x^3 + 3x^2y - y^3$.

Exercise 5.2 [Easy]
Compute the Laplacian Δf of $f(x, y, z) = x^2 + y^2 + z^2$.

Exercise 5.3 [Medium]
Let $f(x, y) = e^{xy}$. Compute the full Hessian H_f and verify Schwarz's theorem by checking $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$.

Exercise 5.4 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be given by $f(x) = \frac{1}{2} x^T A x$ where $A \in \mathbb{R}^{n \times n}$ is a constant matrix. Show that $H_f(x) = \frac{1}{2}(A + A^T)$ for all x. What happens if A is symmetric?

Directional Derivatives and Steepest Ascent

Exercise 6.1 [Easy]
Let $f(x, y) = x^2 - xy + y^2$. Compute the directional derivative of f at the point $(1, 2)$ in the direction $v = \frac{1}{\sqrt{2}}(1, 1)^T$.

Exercise 6.2 [Medium]
Let $f(x, y) = e^{-(x^2+4y^2)}$. At the point $(1, 0)$:

(a) Find the direction of steepest ascent (using the Euclidean norm).

(b) Find the maximum rate of increase.

Exercise 6.3 [Medium]
Let $f(x, y) = x^2 + y^2$ and suppose we measure the direction v in the ℓ_∞ norm, i.e., $\|v\|_\infty = 1$. Find the direction of maximal ascent at the point $(3, 1)$.

Hint: The direction of maximal ascent under $\|v\|_\infty = 1$ is $v_i = \text{sign}((\nabla f)_i)$.

Exercise 6.4 [Hard]
Prove that among all unit vectors v with $\|v\|_2 = 1$, the directional derivative $D_v f(x)$ is maximized when $v = \frac{\nabla f(x)}{\|\nabla f(x)\|_2}$ (assuming $\nabla f(x) \neq 0$), and that the maximum value is $\|\nabla f(x)\|_2$.

Hint: Use the Cauchy–Schwarz inequality.

Level Sets and Gradient Orthogonality

Exercise 7.1 [Easy]
Let $f(x, y) = x^2 + 4y^2$. Sketch (or describe) the level set $L_f(4) = \{(x, y) : f(x, y) = 4\}$. What is ∇f at the point $(2, 0)$? Verify that it is orthogonal to the level curve at that point.

Exercise 7.2 [Medium]
Let $f \in C^1(\mathbb{R}^n)$ and let $\gamma : [0, 1] \to \mathbb{R}^n$ be a curve contained in the level set $L_f(c) = \{x : f(x) = c\}$. Prove that $\nabla f(\gamma(t))$ is orthogonal to $\dot{\gamma}(t)$ for all $t \in [0, 1]$.

Mean Value Theorem and Lipschitz Continuity

Exercise 8.1 [Easy]
Let $f(x, y) = x^2 + y^2$ defined on $\Omega = \{(x, y) : x^2 + y^2 \leq 1\}$ (the closed unit disk). Find an upper bound for $|f(x) - f(y)|$ for $x, y \in \Omega$ using the gradient bound.

Hint: Bound $\|\nabla f(z)\|_2$ on Ω.

Exercise 8.2 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be in $C^1(\mathbb{R}^n)$ and suppose $\|\nabla f(x)\|_2 \leq L$ for all $x \in \mathbb{R}^n$. Using the fundamental theorem of calculus along the straight-line path $\gamma(t) = x + t(y - x)$, prove that $|f(y) - f(x)| \leq L\|y - x\|_2$.

Exercise 8.3 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be in $C^1(\mathbb{R}^n)$. Suppose f is L-Lipschitz continuous: $|f(x) - f(y)| \leq L\|x - y\|_2$ for all x, y. Prove that $\|\nabla f(x)\|_2 \leq L$ for all x.

Hint: Consider $y = x + tv$ for unit vector v and take $t \to 0$.

Mixed Challenging Exercises

Exercise 9.1 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable. Show that the Hessian $H_f(x)$ satisfies:

$$H_f(x) = \lim_{h \to 0} \frac{f(x + h) - f(x) - \langle \nabla f(x), h \rangle - \frac{1}{2} h^T H_f(x) h}{\|h\|^2} = 0$$

by writing the second-order Taylor expansion of f around x.

Exercise 9.2 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(x) = \|Ax - b\|_2^2$ where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Compute $\nabla f(x)$ and $H_f(x)$. Use these to find the critical points of f.

Exercise 9.3 [Hard]

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by $f(x, y) = e^{-\langle z, Bz \rangle}$ where $z = (x, y)^T$ and $B = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$.

(a) Compute $\nabla f(x, y)$ using the chain rule.

(b) Show that ∇f is orthogonal to the level sets of f.

(c) Is f Lipschitz continuous on $\mathbb{R}^2$? Justify your answer.

Exercises: Multivariate Taylor Expansion, Critical Points & Extrema

Mathematical Foundations for Machine Learning

Taylor Expansion — Computation

Exercise 1.1 [Easy]
Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined as $f(x, y) = e^{x+y}$. Compute the second-order Taylor expansion of f around $x_0 = (0, 0)$.

Exercise 1.2 [Easy]
Let $f(x, y) = \sin(x)\cos(y)$. Compute the second-order Taylor expansion around $(0, 0)$.

Exercise 1.3 [Medium]
Given $f : \mathbb{R}^n \to \mathbb{R}$ defined by $f(x) = \sqrt{1 + \|x\|^2}$. Compute the second-order Taylor expansion at $x_0 = 0$.

(Exam 2026, Exercise 2b)

Exercise 1.4 [Medium]
Let $f : \mathbb{R}^2 \to \mathbb{R}$ with $f(x, y) = \log(1+x^2+y^2+xy)$. Compute the second-order Taylor expansion around $(0, 0)$ and verify your result by comparing with the expansion of $\log(1 + u)$ for small u.

Exercise 1.5 [Hard]
Let $A \in \mathbb{R}^{n\times n}$ be symmetric and positive definite. Define $f : \mathbb{R}^n \to \mathbb{R}$ by $f(x) = \log(x^T Ax)$. Compute the second-order Taylor expansion around any point x_0 with $x_0^T Ax_0 \neq 0$.

Critical Points — Finding and Classifying

Exercise 2.1 [Easy]
Find all critical points of $f(x, y) = x^2 + y^2 - 2x - 4y + 5$ and classify them as local min, local max, or saddle point.

Exercise 2.2 [Easy]
Find and classify all critical points of $f(x, y) = x^3 - 3xy + y^3$.

Exercise 2.3 [Medium]
Find and classify all critical points of $f(x, y) = (x^2 + y^2)e^{-(x^2+y^2)}$.

-> lots of computation, only to practice algebra, the one below better for recall

Exercise 2.4 [Medium]
Consider the function from Exam 2021 Exercise 2a in a related form: let $f(x, y) = x^2 + xy + y^2 - 3x - 3y$. Find all critical points, classify them, and compute the value of f at each.

Then compute the directional derivative at the critical point in the direction $v = \begin{pmatrix}1\\2\end{pmatrix}$ and explain why the result makes sense.

no first order change != no change

Exercise 2.5 [Hard]
Consider $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x, y) = x^2 + y^2(1 - x)^3$.

(a) Find all critical points and classify them using the Hessian.

(b) Show that f has no global minimum by finding a direction along which $f \to -\infty$.

(c) Does this contradict the result that, for $f \in C^1(\mathbb{R})$, a unique critical point that is a local minimum must be the global minimum? Explain carefully.

Local vs. Global Extrema

When hessian eigenvalues are 0 that means that the second grade curvature is 0. But it could be that the higher order curvature be neg or pos. (when we have second order curvature it shadows the higher order curvature, because since we do taylor, we are talking about a tiny change, 0.00x, so very tiny^2 > tiny^3, 4...

Exercise 3.1 [Easy]
State the necessary condition and the sufficient condition for a critical point x_0 of $f \in C^2(\Omega)$ to be a local minimum. Give a concrete example where the necessary condition is satisfied but the sufficient condition is not.

Exercise 3.2 [Medium]
Let $f \in C^1(\mathbb{R})$ have a single critical point x_0 which is a local minimum. Prove that x_0 is the global minimum.

Hint: Use the theorem of Rolle by contradiction.
(Exam 2021 style — proof question)

Exercise 3.3 [Medium]
Prove that any local minimum of a convex function $f : \Omega \to \mathbb{R}$ over a convex set Ω is also a global minimum.

Hint: Prove by contradiction using the definition of convexity.
(Exam 2021, Exercise 3a)

Exercise 3.4 [Hard]
Let $f \in C^1(\mathbb{R}^n)$ have a unique critical point x_0 which is a local minimum. Suppose additionally that there exists $r > 0$ such that $f(x) > f(x_0)$ for all $x \in \mathbb{R}^n \setminus B(x_0, r)$. Prove that x_0 is the global minimum.

Hint: Use compactness of the closed ball and the extreme value theorem.

Positive Semi-Definite Hessians and Degenerate Cases

Exercise 4.1 [Easy]
Classify the critical point at the origin for each of the following functions:

(a) $f(x, y) = x^2 + y^4$

(b) $f(x, y) = x^2 - y^4$

(c) $f(x, y) = x^4 + y^4$

Note that the Hessian is only semi-definite at the origin for each. Can the Hessian test conclude anything? Determine the nature of each critical point from the definition.

Exercise 4.2 [Medium]
Let $\|u\| = 1$ and define $f : \mathbb{R}^n \to \mathbb{R}$ by

$$f(x) = \langle x, (I - uu^T)x \rangle - \langle u, x \rangle^4.$$

Show that $\nabla f(0) = 0$ and $Hf(0) \succeq 0$, but $x_0 = 0$ is **not** a local minimum.
(This is the counterexample from the lecture notes.)

To prove one direction then take alfa * u

Exercise 4.3 [Hard]
Let $f \in C^2(\mathbb{R}^n)$ with $\nabla f(x_0) = 0$. Prove the following:

(a) (Necessary condition) If x_0 is a local minimum, then $Hf(x_0) \succeq 0$.

(b) (Sufficient condition) If $Hf(x_0) \succ 0$, then x_0 is a strict local minimum.

Use the second-order Taylor expansion with remainder.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
Let $W \in \mathbb{R}^{m\times n}$ and define the loss function of a single-layer linear network:

$$L(W) = \|y - Wx\|_2^2$$

where $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$ are fixed.

(a) Compute $\nabla_W L$ and find all critical points.

(b) Compute the Hessian. *Hint:* treat W as a vector in $\mathbb{R}^{mn}$ via $\mathrm{vec}(W)$, and use the identity $\mathrm{vec}(ABC) = (C^T \otimes A)\mathrm{vec}(B)$.

(c) Classify the critical point(s). Under what condition on x is the critical point a strict local minimum?

Exercise 5.2 [Hard]
Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive definite. Consider the function $f : \mathbb{R}^n \to \mathbb{R}$:

$$f(x) = \frac{1}{2}x^T Ax - b^T x$$

where $b \in \mathbb{R}^n$ is fixed.

(a) Find the unique critical point and prove it is a strict global minimum.

(b) Compute the second-order Taylor expansion of f around the minimizer x^*.

(c) Prove that $f(x) - f(x^*) = \frac{1}{2}(x - x^*)^T A(x - x^*)$ for all x. Interpret this geometrically in terms of the level sets of f.

Exercise 5.3 [Hard]
Let $f \in C^2(\mathbb{R}^n)$ and suppose x_0 is a critical point with $Hf(x_0)$ having eigenvalues $\lambda_1 < 0 < \lambda_2$, with corresponding eigenvectors u_1, u_2. Using the Taylor expansion:

(a) Prove that x_0 is a saddle point by showing that f increases along one direction and decreases along another.

(b) In the context of training a neural network with loss function $L(\theta)$, explain why a saddle point might be more problematic than a local minimum for gradient descent. How does the second-order information from Hf help?

Exercises: Constrained Optimization and the Implicit Function Theorem

Mathematical Foundations for Machine Learning

If we are in f:Rp×RmRm then we can write every m input varin m output v

Implicit Function Theorem

Exercise 1.1 [Easy]
Consider the equation $F(x, y) = x^2 + y^2 - 1 = 0$ and the point $(x_0, y_0) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.

(a) Verify that $F(x_0, y_0) = 0$.

(b) Check the hypothesis of the Implicit Function Theorem: compute $\frac{\partial F}{\partial y}(x_0, y_0)$ and verify it is nonzero.

(c) Conclude that there exists a function $y = \varphi(x)$ defined near x_0 such that $F(x, \varphi(x)) = 0$. Find $\varphi'(x_0)$ by implicit differentiation.

From that we can reconstruct taylor because we get '(x)y(x0)

Exercise 1.2 [Medium]
Consider the system of equations $F : \mathbb{R}^3 \to \mathbb{R}$ defined by

$$F(x_1, x_2, x_3) = x_1^2 - x_1 x_3 + x_2^2 x_1 - x_2^2 = 0.$$

(a) Verify that $z = (1, 1, 1)$ is a solution.

(b) Can we locally solve for x_3 as a function of (x_1, x_2) near z? Justify using the Implicit Function Theorem.

(c) Compute $\frac{\partial x_3}{\partial x_1}$ and $\frac{\partial x_3}{\partial x_2}$ at the point $(1, 1)$ by implicit differentiation.

(d) At what points does the IFT fail to guarantee a local solution for x_3? Interpret geometrically.

Exercise 1.3 [Hard]
Let $F : \mathbb{R}^4 \to \mathbb{R}^2$ be defined by

$$F(x, y, u, v) = \begin{pmatrix} x^2 + y^2 - u^2 - v \\ xy - uv \end{pmatrix}.$$

Consider the point $z_0 = (1, 1, 1, 1)$.

(a) Verify that $F(z_0) = 0$.

(b) Can we locally solve for (u, v) as functions of (x, y) near z_0? Set up and verify the appropriate condition from the Implicit Function Theorem.

(c) Compute the Jacobian matrix $D_{(x,y)}(u, v)$ at $(1, 1)$ using the implicit differentiation formula:

$$D_y \varphi = -(D_y F)^{-1} \cdot D_x F,$$

where $\varphi(x, y) = (u(x, y),\ v(x, y))$.

Lagrange Multipliers — Setup and Computation

Exercise 2.1 [Easy]
Find the extrema of $f(x, y) = x + y$ subject to the constraint $x^2 + y^2 = 1$.

(a) Write down the Lagrangian $\mathcal{L}(x, y, \lambda)$.

(b) Compute the stationarity conditions $\nabla_{x,y}\mathcal{L} = 0$ and the constraint equation.

(c) Solve the system to find all critical points and classify them as maxima or minima.

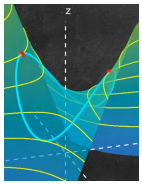
The gradient of the restriction g and the function to minimize f are parallel to each other and perpendicular to the tangent line.

Exercise 2.2 [Easy]
Find the point on the plane $2x + y + z = 5$ that is closest to the origin.

(a) Formulate this as a constrained optimization problem: state the objective function and the constraint.

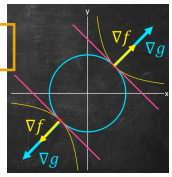
(b) Write the Lagrangian and derive the stationarity conditions.

(c) Solve for the closest point.



f(c)=g(c)

c is one of the two opt



Exercise 2.3 [Medium]
Consider the curve $y = 1.5 - x^2$ in $\mathbb{R}^2$.

(a) Formulate the problem of finding the point(s) on this curve closest to the origin as a constrained optimization problem using the squared Euclidean distance.

(b) Write down the Lagrangian. Derive the stationarity conditions.

(c) Solve the system. You should find that there are two cases depending on the value of λ. Find all constrained minimizers.

(d) State the necessary and sufficient conditions for a point to be a constrained minimizer in general.

We take the direction in which we fulfill the constraints: v,g(c)=0The hessiar

Exercise 2.4 [Medium]
Find the closest point to $(0, 0)$ in the set

$$M = \{x \in \mathbb{R}^2 : x_1 + x_2 \geq 4,\ 2x_1 + x_2 \geq 5\}.$$

(a) Formulate as an optimization problem with inequality constraints. Write the objective and constraints in the standard form $g_i(x) \leq 0$.

(b) Write the Lagrangian and state all four KKT conditions.

(c) Solve the KKT system. Use complementary slackness to enumerate cases.

Exercise 2.5 [Medium]
Consider the entropy maximization problem on the probability simplex:

$$\max_{x \in \mathbb{R}^n} \left(-\sum_{i=1}^{n} x_i \log x_i \right) \quad \text{s.t.} \quad \sum_{i=1}^{n} x_i = 1, \quad x_i \geq 0.$$

(a) Write the Lagrangian incorporating both the equality and inequality constraints.

(b) Derive the stationarity condition with respect to x_i.

(c) Using complementary slackness, argue that $x_i > 0$ for all i at the optimum. Then solve for x_i.

(d) What is the maximum entropy value?

Lagrange Multipliers — Proofs and Theory

Exercise 3.1 [Easy]
Let $g : \mathbb{R}^n \to \mathbb{R}$ be differentiable and $N = \{x : g(x) = 0\}$. Suppose $c \in N$ is a local minimum of a differentiable function f on N, and $\nabla g(c) \neq 0$.

(a) State the necessary condition (Lagrange condition) that must hold at c.

(b) Give a geometric interpretation: what does $\nabla f(c) = \lambda \nabla g(c)$ mean about the gradients at the constrained optimum?

(c) Why is the condition $\nabla g(c) \neq 0$ (constraint qualification) needed?

Exercise 3.2 [Medium]
Proving the Lagrange necessary condition in $\mathbb{R}^2$.

Let $f, g : \mathbb{R}^2 \to \mathbb{R}$ be C^1 functions, $N = \{(x, y) : g(x, y) = 0\}$, and suppose $c = (a, b) \in N$ is a local minimum of f on N with $\frac{\partial g}{\partial y}(c) \neq 0$.

(a) By the Implicit Function Theorem, there exists φ with $g(x, \varphi(x)) = 0$ near a. Define $h(x) = f(x, \varphi(x))$. Explain why $h'(a) = 0$.

(b) Compute $h'(a)$ using the chain rule and implicit differentiation (express $\varphi'(a)$ in terms of partial derivatives of g).

(c) Use parts (a) and (b) to show that $\nabla f(c) = \lambda \nabla g(c)$ for some $\lambda \in \mathbb{R}$.

Exercise 3.3 [Hard]
Sufficient condition for a constrained minimum.

Let $f, g : \mathbb{R}^n \to \mathbb{R}$ be C^2, $N = \{x : g(x) = 0\}$, and suppose at $c \in N$ there exists λ such that the Lagrangian $h = f - \lambda g$ satisfies $\nabla h(c) = 0$.

Define the **constrained tangent space** $T_c = \{v \in \mathbb{R}^n : \langle v, \nabla g(c) \rangle = 0\}$.

(a) State the sufficient condition for c to be a strict local constrained minimum in terms of the Hessian of h restricted to T_c.

(b) *Application.* Consider $f(x, y) = x^2 + 2y^2$ subject to $g(x, y) = x + y - 1 = 0$. Find the critical point using Lagrange multipliers. Then verify the sufficient condition to confirm it is a local minimum.

(c) *Application.* Consider maximizing $f(x, y, z) = xyz$ on the sphere $x^2 + y^2 + z^2 = 1$. Find the critical points in the first octant $(x, y, z > 0)$. Verify using the bordered Hessian or the restricted Hessian that this is indeed a local maximum.

Lagrange Multipliers with Inequality Constraints (KKT)

Exercise 4.1 [Easy]
State all four Karush–Kuhn–Tucker (KKT) conditions for the problem

$$\min_x f(x) \quad \text{s.t.} \quad g_i(x) \leq 0,\ i = 1, \ldots, m, \quad h_j(x) = 0,\ j = 1, \ldots, p.$$

Briefly explain the role of each condition.

Exercise 4.2 [Medium]
Minimize $f(x_1, x_2) = (x_1 - 3)^2 + (x_2 - 2)^2$ subject to

$$x_1 + x_2 \leq 4, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

(a) Rewrite the constraints in standard form $g_i(x) \leq 0$.

(b) Write the Lagrangian and the full set of KKT conditions.

(c) Solve the system. Enumerate cases using complementary slackness and find the optimal solution.

Exercise 4.3 [Hard]
Consider the KL-divergence minimization problem from the 2026 exam:

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^n x_i (\log x_i - \log p_i) \quad \text{s.t.} \quad \sum_{i=1}^n x_i = 1, \quad x_i \geq 0,$$

where $p_i > 0$ are given with $\sum_i p_i = 1$.

(a) Is the objective function convex? Justify.

(b) Is the objective function **strongly** convex? Prove or disprove.

(c) Assuming strong duality holds, write down the Lagrangian, the dual function (minimizing over x), and solve the dual problem.

(d) What is the optimal solution x^*? Interpret the result in terms of information theory.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
Spectral optimization via Lagrange multipliers.

Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix with eigenvalues $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$.

(a) Formulate the problem $\max_{\|x\|_2=1} x^T Ax$ as a constrained optimization problem and apply the method of Lagrange multipliers. Show that the critical points satisfy $Ax = \mu x$ for some multiplier μ, i.e., they are eigenvectors.

(b) Conclude that the maximum of the Rayleigh quotient $R_A(x) = \frac{x^T Ax}{x^T x}$ equals λ_n.

(c) Now consider the 2026 exam-style problem: $A = aa^T + bb^T$ with $a = u + v$, $b = \frac{3}{4}u - 3v$, where u, v are orthonormal. Find $\max_{\|x\|=1} \|Ax\|_2 - \langle x, Ax\rangle$.

(*Hint: First compute the eigenvalues of A, then express the objective in spectral coordinates.*)

Exercise 5.2 [Hard]
From constrained optimization to MLE.

Consider N data points $\{(x_n, y_n)\}_{n=1}^N$ with $x_n \in \mathbb{R}^D$, $y_n \in \mathbb{R}$, arising from $y = x^T\theta + \varepsilon$ where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ i.i.d.

(a) Write the conditional probability $P(Y|X;\theta)$ using the i.i.d. assumption.

(b) Show that the maximum likelihood estimate is $\hat{\theta} = (X^T X)^{-1} X^T Y$.

(c) Now suppose we add a regularization constraint $\|\theta\|_2^2 \le r^2$ (Ridge-style). Formulate the constrained MLE as a Lagrangian problem and use KKT conditions to show that the solution satisfies

$$\hat{\theta}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T Y$$

for some $\lambda \ge 0$.

(d) Interpret the role of λ: what happens when $\lambda = 0$ vs. $\lambda \to \infty$?

Exercise 5.3 [Hard]
Lagrange multipliers meet the Implicit Function Theorem.

Let $f, g : \mathbb{R}^3 \to \mathbb{R}$ be C^2 functions. Define the Lagrangian system $F : \mathbb{R}^4 \to \mathbb{R}^3$ by

$$F(x_1, x_2, x_3, \lambda) = \begin{pmatrix} \nabla f(x) - \lambda \nabla g(x) \\ g(x) \end{pmatrix} = \begin{pmatrix} \partial_1 f - \lambda \partial_1 g \\ \partial_2 f - \lambda \partial_2 g \\ \partial_3 f - \lambda \partial_3 g \\ g(x) \end{pmatrix} \in \mathbb{R}^4.$$

(Note: F maps $\mathbb{R}^4 \to \mathbb{R}^4$.)

Suppose (x^*, λ^*) is a solution of $F = 0$.

(a) Under what conditions on f and g can we apply the Implicit Function Theorem to conclude that the solution (x^*, λ^*) is locally unique (isolated)?

(b) Compute the Jacobian DF explicitly in terms of the Hessians of f and g and the gradient ∇g.

(c) *Specific case.* Let $f(x) = x_1^2 + x_2^2 + x_3^2$ and $g(x) = x_1 + x_2 + x_3 - 3 = 0$. Find the constrained minimum using Lagrange multipliers and verify that the Jacobian DF is invertible at the solution.